

JWST/NIRCam F150W Surface Brightness Fluctuation Distances from GO-3055: A TRGB-Anchored Calibration and Internal Accuracy Test

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Abstract

We measure JWST/NIRCam F150W surface brightness fluctuation (SBF) distances for 14 early-type galaxies in program GO-3055 and calibrate them directly with the individual JWST tip of the red giant branch (TRGB) distances from Chazov et al. (2026). The primary experiment is fixed in advance: all 14 galaxies, individual Paper IV thin-slice distances, no manual rejection, and one color-independent F150W zero point. A linear color term is tested before the distance analysis, but is not adopted because it does not improve either the information criterion or cross-validation. The resulting calibration is $\overline{M}_{150,0} = -3.208 \pm 0.024_{\text{stat}} \pm 0.051_{\text{common}}$ mag, with fitted intrinsic scatter 0.081 mag (bootstrap 16th–84th percentile interval 0.064–0.088 mag). Its leave-one-out (LOO) distance-modulus bias is -0.0010 mag and its RMS is 0.0925 mag, corresponding to 4.3 per cent in distance. The median predicted internal uncertainty is 0.0865 mag (4.1 per cent); including the common TRGB absolute-scale uncertainty gives 0.0984 mag (4.6 per cent). The normalized LOO residuals have RMS 1.08; eight of 14 objects are within 1σ and all 14 are within 2σ . For comparison, a linear color law gives a worse LOO RMS of 0.0981 mag, is disfavored by $\Delta\text{AICc} = 3.15$, and has a bootstrap slope interval that includes zero. The data therefore do not require a color dependence, and all primary distances and uncertainties below use the constant zero point. Paper IV thick-slice distances and the Paper III cluster-mean prescription give consistent accuracy. The result is an internal cross-validation of the calibrators, not an independent external validation.

Keywords: distance scale; galaxies; surface brightness fluctuations; JWST; NIRCam; TRGB.

1 Introduction

Surface brightness fluctuations arise from Poisson variations in the number and luminosity of unresolved stars within each resolution element [1]. Their apparent amplitude decreases with distance, so an absolute SBF magnitude calibrated for the stellar population provides a distance indicator. In the near-infrared, evolved stars produce a strong SBF signal, but the absolute amplitude may depend on age and metallicity. Color is therefore commonly used as a population correction; it is an op-

tional improvement to the distance estimator, not the definition of the method.

The TRGB-SBF project is constructing a Population-II distance ladder that transfers a geometrically calibrated TRGB zero point to SBF. Jensen et al. [Paper III; 13] used JWST TRGB distances to recalibrate the existing HST/WFC3-IR F110W SBF scale. With mean Virgo and Fornax distances, they obtained

$$\overline{M}_{110} = (1.86 \pm 0.16)[(g - z)_{\text{PS}} - 1.30] - (2.760 \pm 0.016).$$

Here the SBF signal is measured from the JWST GO-3055 F150W images. The F090W–F150W color is measured independently only to test whether a population correction is required; it is not an input to the F150W power-spectrum measurement. The anchors are the revised JWST TRGB distances of Chazov et al. [Paper IV; 14], tied to the NGC 4258 maser distance. Our main questions are deliberately ordered as follows: (1) is a color term required; (2) if not, how accurately does a single F150W SBF zero point recover the TRGB distances; and (3) is the propagated uncertainty consistent with the observed residuals?

2 Data and distance anchors

2.1 GO-3055 sample

The sample contains 14 galaxies: three Fornax members, nine galaxies in or near the Virgo region, and two nearby group galaxies. In tables and plotting legends, “Virgo” or “Virgo region” is only this broad sky/environment class; it is not a claim of strict Virgo Cluster membership. In particular, NGC 4636 and foreground NGC 4697 are excluded from the strict seven-object Virgo Cluster subset. Every target has NIRCam F090W and F150W imaging. The signal and color are measured in the same two circular annuli, 8.2–16.4 and 16.4–32.8 arcsec, following the geometry of Paper III. At the individual Paper IV distances, the outer 32.8-arcsec radius spans 1.65–3.17 kpc. Consequently, radial differences cannot be interpreted as measurements at a common physical radius. Jensen et al. also used an auxiliary elliptical-annulus check for elongated galaxies; that secondary test is not reproduced here, and the limitation is kept explicit.

The primary anchors are the individual Paper IV thin-slice TRGB moduli. This choice was fixed before inspection of the present SBF residuals and is kept to avoid selecting an anchor variant after seeing which gives less scatter. Paper IV prefers its thick-slice estimator because it uses a broader and more robust stellar sample on the blue side of the TRGB color break. We therefore treat thick-slice distances as a mandatory sensitivity experiment rather than silently ignoring that preference.

Two Jensen comparisons are kept separate throughout. Paper III is the HST F110W calibration of 14 TRGB–SBF targets and is used for the color-law and cluster-mean comparisons. A legacy Jensen et al. (2015) F160W reconstruction exists for only five common galaxies and is reported only as a separately labelled distance column; it is never merged with the Paper III F110W sample.

The extinction-corrected measurements, calibration-point uncertainties, and LOO residuals are reported together in Table 1 only after the measurement and statistical procedures have been defined. This keeps the data description separate from the derived results.

2.2 Foreground extinction

Paper IV tabulates A_{090} . Using its coefficients, we adopt

$$E(B - V) = A_{090}/1.4156, \quad (1)$$

$$A_{150} = 0.6021E(B - V), \quad (2)$$

$$C_0 = C_{\text{obs}} - (A_{090} - A_{150}), \quad (3)$$

$$\bar{m}_{150,0} = \bar{m}_{150,\text{obs}} - A_{150}. \quad (4)$$

The corrections make the colors bluer by 0.008–0.029 mag and the F150W SBF magnitudes brighter by 0.006–0.022 mag. They do not materially change the distance RMS, but they are required for an unbiased zero point and covariance matrix.

No separate internal-extinction correction is applied. The central 8.2 arcsec are excluded, obvious non-elliptical features enter the object mask, and the sample contains no retained strong dust lane. As a quantitative sensitivity bound, an unrecognized screen with $A_V^{\text{int}} \leq 0.05$ mag would give $A_{150}^{\text{int}} \lesssim 0.01$ mag for a standard near-infrared extinction curve, well below the observed 0.0925-mag LOO scatter. This is a bound, not a measurement of internal dust.

All targets have $z < 0.007$. For a deliberately broad local spectral-slope range $|1 + \alpha| \leq 3$, where $f_\nu \propto \nu^\alpha$, the ordinary same-filter correction satisfies

$$|K_{150}| = 2.5|1 + \alpha|\log_{10}(1 + z) < 0.023 \text{ mag}. \quad (5)$$

It is not applied object by object because a consistent SBF spectral-energy model is unavailable and the bound is subdominant at the present precision. Likewise, cosmological surface-brightness dimming is not added as a separate SBF term: the common surface-brightness factor cancels in the variance divided by mean-galaxy-light normalization. Even the full Tolman term is bounded by $10\log_{10}(1 + z) < 0.030$ mag over the sample. These two bounds delimit, rather than conceal, the small-redshift approximation.

3 SBF measurement

3.1 Sky, galaxy model, and masks

The input images are the public calibrated NIRC*am* i2*d* mosaics in MJy sr^{−1}. A scalar sky is estimated outside the science annuli from independent background regions and subtracted. Invalid pixels, detector gaps, foreground objects, and compact sources form the primary mask. Initial compact sources require at least nine connected pixels above a local 3.5σ threshold; after galaxy subtraction, a second catalogue uses four connected pixels above 2.5σ , deblending, and a three-pixel morphological dilation. The mask is deliberately aggressive toward compact positive residuals while retaining diffuse galaxy structure.

The centre is initialized from the image and checked against the WCS target position. Ellipticity and position angle remain free while isophotes are fitted from a starting semimajor axis of 50 pixels to at most 2000 pixels. Intensities and fitted geometry are linearly interpolated between consecutive isophotes; there is no piecewise constant filling between isophotal lines. A Ser-sic image is used only to fill detector gaps while solving the isophotes. The final model at every real science pixel comes from the measured isophotal profile. Filled gap pixels never enter a power spectrum.

After sky and galaxy-model subtraction, the residual is first divided by $\sqrt{M_i}$. The centre and scale of this *normalized* field are then estimated with up to five iterations of symmetric 3.5σ -clipped statistics, using every catalogue-unmasked pixel for which the science image is finite and $M_i > 0$. This support is the complete valid positive-model region: it is not the raw-residual distribution and not the union of the two measurement annuli. Values outside the resulting median plus or minus 3.5 scale interval are replaced by the corresponding boundary value. Pixels are not rejected. This operation is therefore *winsorization*, not sigma-clipping; sigma clipping is used only to estimate the centre and scale.

The resulting full-frame normalized and winsorized image is saved as a FITS product before any annulus is selected. Pixels rejected by the catalogue mask, invalid science pixels, and pixels with a non-positive model remain NaN in this product and never enter either the threshold estimate or a Fourier spectrum. The previous raw-residual and annulus-union prescriptions are retained only as comparison branches. Robustification is a deliberate implementation change relative to Jensen and is therefore tested explicitly rather than presented as part of the reproduced method.

Figure 1 shows these saved full-frame residual products. Masked pixels are plotted in black; they are absent from both $P(k)$ and $E(k)$ rather than being assigned a numerical residual value. The mean of each selected annulus is removed only when its exact FFT input is constructed, so that mean subtraction is not baked into the saved full-frame FITS.

Final normalized F150W FFT inputs: full-support 3.5σ winsorization

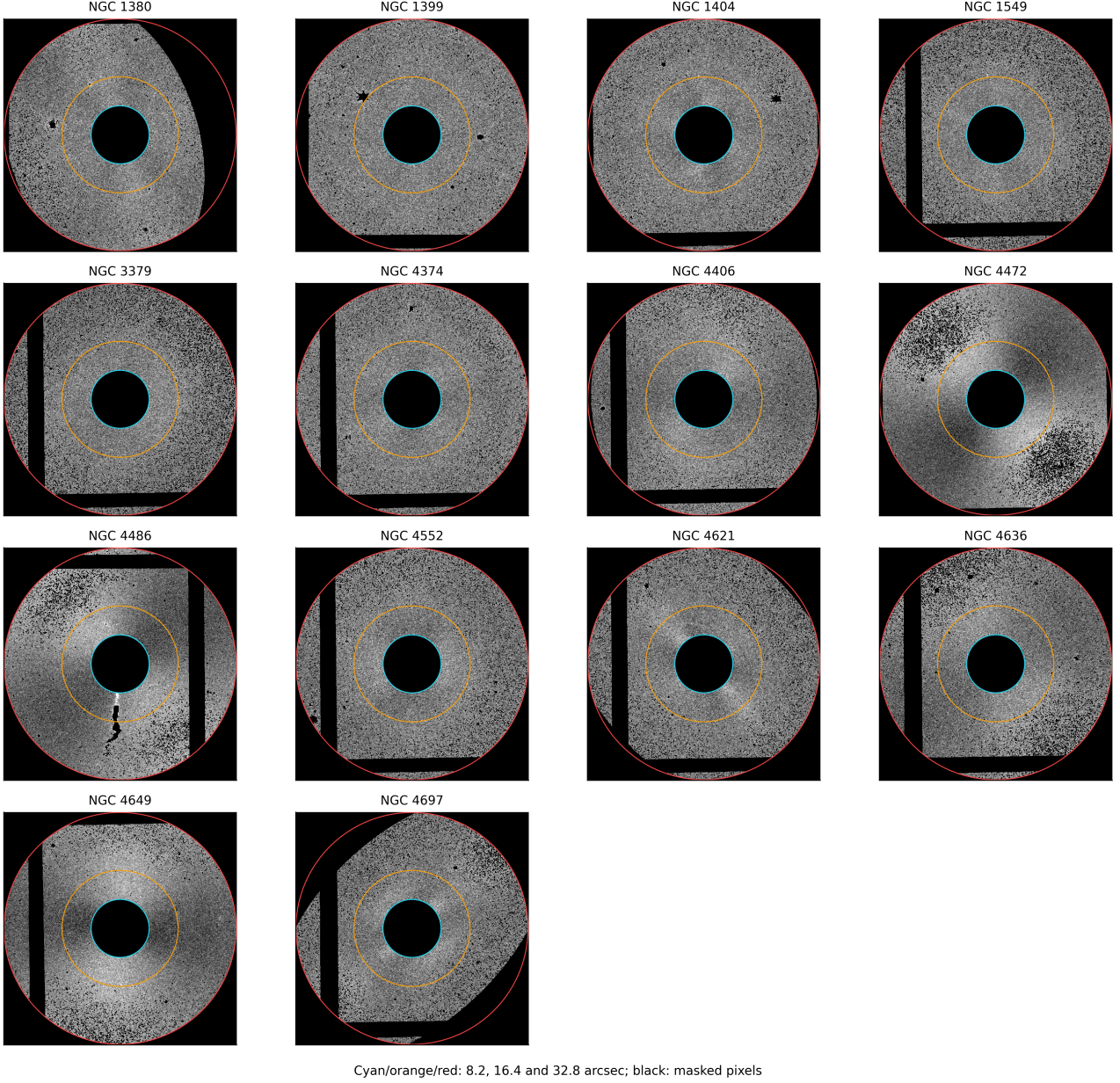


Figure 1: Full-frame F150W residual products after division by $\sqrt{M_i}$ and symmetric 3.5σ winsorization. The smooth galaxy model has been subtracted and compact sources and invalid image regions are masked. Black pixels are excluded from the analysis; visible grey-scale structure in the unmasked area is real residual structure, not a mask value. The saved FITS still contains the full valid support and has not been mean-subtracted. An annulus is selected and its mean is removed only immediately before its FFT is calculated.

3.2 Normalized residual and expectation spectrum

For a smooth model M_i and sky-subtracted data I_i , define the normalized residual

$$Q_i = \frac{I_i - M_i}{\sqrt{M_i}}. \quad (6)$$

The division by $\sqrt{M_i}$ converts the Poisson stellar variance, which is proportional to local galaxy intensity, into an approximately stationary SBF amplitude. It does not measure a difference between the brightest and faintest pixels. SBF is the variance of the unresolved stellar luminosity function per unit mean light, recovered statistically from all usable pixels.

Let Q_i^{win} be Q_i after applying the single full-support winsorization interval described above, and let W_i be the binary window of one selected annulus. For that annulus the exact field entering the Fourier analysis is

$$R_i = W_i (Q_i^{\text{win}} - \langle Q^{\text{win}} \rangle_W). \quad (7)$$

Thus the threshold is estimated once over the full valid image, whereas the mean is estimated separately after selecting each ring. NaN pixels in the saved FITS are absent from the mean and from the FFT; zeros outside W_i in the explicit FFT array represent multiplication by the binary window, not measured zero-valued residuals. We calculate the two-dimensional FFT and use

$$P(\mathbf{k}) = \frac{|\mathcal{F}[R](\mathbf{k})|^2}{N_{\text{use}}}. \quad (8)$$

The spectrum is azimuthally averaged into 80 radial bins; the uncertainty of each bin is the standard error of its contributing Fourier modes. The expected SBF spectrum is not simply $|\text{PSF}|^2$. For each of five time- and position-matched, unit-normalized STPSF models, 64 independent unit white-noise fields are convolved with the PSF, multiplied by the exact annular window and source mask, mean-subtracted, Fourier transformed, and averaged:

$$E_j(k) = \left\langle \frac{|\mathcal{F}\{W[n * \text{PSF}_j]\}|^2}{N_{\text{use}}} \right\rangle_{64}. \quad (9)$$

Thus $E(k)$ includes PSF blurring, the finite annulus, holes in the source mask, and mode mixing by the window.

The same N_{use}^{-1} Fourier normalization is applied to the measured field and every Monte Carlo realization used to construct $E(k)$, and every PSF has unit summed flux before convolution. Therefore the fitted coefficient P_0 is already on the fluctuation-flux scale defined by Eq. 7. No additional division by the mean model brightness is permitted after the fit; doing so would normalize the galaxy light twice. In the diagnostic power-spectrum panels, the black points are the azimuthal means of the measured two-dimensional power spectrum in radial Fourier bins, and their error bars are the standard errors of the contributing Fourier modes. They are not individual image pixels or PSF samples.

3.3 Power-spectrum fit and fluctuation magnitude

For every PSF realization we fit

$$P(k) = P_{0,j}E_j(k) + P_{1,j} \quad (10)$$

by weighted least squares, with weights equal to the inverse squared bin errors. The covariance is $(X^T W X)^{-1}$ multiplied by $\max(\chi_\nu^2, 1)$. The adopted P_0 and P_1 are the medians over the five PSFs. The primary normalized-frequency interval is $0.04 \leq k \leq 0.25$, with an additional lower bound of ten Fourier modes across the shorter crop axis. Fits beginning at $k_{\text{min}} = 0.01$ and 0.03 , with the same k_{max} , quantify low-frequency sensitivity but do not replace the primary fit.

The PSF-shaped power contains both stellar SBF and unresolved compact sources:

$$P_{\text{SBF}} = P_0 - P_r. \quad (11)$$

Here P_r is specifically the contribution from unmasked globular clusters and background galaxies fainter than the catalogue limit. In each annulus the observed source luminosity function is modelled as a Gaussian globular-cluster luminosity function plus a power-law background-galaxy term. Its faint side is integrated with flux-squared weighting,

$$P_r = \langle M^{-1} \rangle \int_{m_{\text{cut}}}^{m_{\text{cut}}+12} [\phi_{\text{GC}}(m) + \phi_{\text{bg}}(m)] f^2(m) dm, \quad (12)$$

where $f(m)$ is the flux in image units and $\langle M^{-1} \rangle$ restores the normalization of Eq. 7. The adopted GCLF width is 1.2 mag; widths of 1.35 mag for NGC 1399 and NGC 4472 are tested separately.

Because Eq. 7 already divides the variance by mean galaxy light, P_{SBF} has units of fluctuation flux and is not divided by M a second time. With pixel solid angle Ω_{pix} ,

$$Z_{\text{AB,pix}} = -2.5 \log_{10} \left(\frac{10^6 \Omega_{\text{pix}}}{3631} \right), \quad (13)$$

$$\bar{m}_{150} = -2.5 \log_{10}(P_0 - P_r) + Z_{\text{AB,pix}}. \quad (14)$$

The inner and outer magnitudes are finally combined with inverse-variance weights. Their half-difference is retained as a radial diagnostic, not inserted as random measurement noise.

All current annular fits give $P_1 < 0$. Corner regions of the original F150W i2d frames have a median logarithmic noise-spectrum slope of -1.45 over the fitted frequency range, so the data are not white. The fitted constant absorbs the local intercept of this misspecified noise spectrum and may be negative; it is not a claim of negative physical noise power.

3.4 Matched color

F090W is registered and PSF-matched to F150W. Color is measured in the same usable pixels and rings as the SBF signal. The two annular colors may differ because the stellar population changes with radius. Their half-difference is therefore retained as a radial diagnostic,

not assigned as a statistical color error. Color does not enter the primary power-spectrum measurement or the adopted distance estimator; it is used only in the model-comparison test of a possible population correction.

4 Uncertainty model

The uncertainty bookkeeping separates errors that vary from galaxy to galaxy, finite-calibration uncertainty, intrinsic population scatter, and common absolute zero points. This separation prevents a common systematic from being incorrectly used to down-weight an individual point.

4.1 Calibration-point errors

Let r denote the inner or outer annulus. The non-PSF power-spectrum error is defined conservatively as

$$\sigma_{k,r} = 1.4826 \text{ MAD}\{\bar{m}_r(k_{\min})\}, \quad (15)$$

$$\sigma_{\text{WLS},r} = \frac{1.085736 \sigma_{P_0, \text{WLS}}}{P_0 - P_r}, \quad (16)$$

$$\sigma_{\text{PS},r} = \max(\sigma_{k,r}, \sigma_{\text{WLS},r}). \quad (17)$$

where the MAD uses $k_{\min} = 0.01, 0.03$, and 0.04 and the second term is the linear propagation of the WLS covariance. It is a maximum, not a quadrature sum, because both terms diagnose uncertainty in the same fitted amplitude. The factor $1.085736 = 2.5/\ln 10$ converts fractional power error to magnitude.

The five PSFs are then kept separate. Their robust scatter is

$$\sigma_{\text{PSF},r} = 1.4826 \text{ MAD}\{\bar{m}_{r,j}\}. \quad (18)$$

The analysis notebook reconstructs Eq. 17 from the saved $\sigma_{P_0, \text{WLS}}$ before adding this term; hence PSF scatter is not counted both inside and outside the fit error. With the same normalized inverse-variance annular weights w_r used for the central magnitude,

$$\sigma_{\text{PS}}^2 = \sum_r w_r^2 \sigma_{\text{PS},r}^2, \quad (19)$$

$$\sigma_{\text{PSF}} = \sum_r w_r \sigma_{\text{PSF},r}. \quad (20)$$

The PSF terms are combined linearly because a single PSF mismatch moves both annuli coherently.

The scalar-sky systematic s is the scatter among independent background estimates in image units. Linearizing Eq. 14 gives

$$\sigma_{\text{sky},r} = 1.085736 \frac{s}{\langle M \rangle_r}, \quad \sigma_{\text{sky}} = \sum_r w_r \sigma_{\text{sky},r}. \quad (21)$$

Again the annuli are correlated because the same scalar sky is subtracted from both. If $q_r = P_r/P_0$, assigning the stated 25 per cent fractional uncertainty to P_r gives

$$\sigma_{P_r,r} = 1.085736 (0.25) \frac{q_r}{1 - q_r}, \quad \sigma_{P_r} = \sum_r w_r \sigma_{P_r,r}. \quad (22)$$

This term is small because $P_r/P_0 < 0.0067$ for every retained annulus; the 0–2 P_r sensitivity experiment tests the stronger possibility that the working luminosity-function model itself is wrong.

The internal apparent-SBF error is therefore

$$\sigma_{\bar{m},\text{int}}^2 = \sigma_{\text{PS}}^2 + \sigma_{\text{PSF}}^2 + \sigma_{\text{sky}}^2 + \sigma_{P_r}^2. \quad (23)$$

The radial half-difference does not appear in Eq. 23: it can be a real population gradient and the two rings share their sky, model, mask, and PSF.

For foreground reddening we assign

$$\sigma_{A_{090}}^2 = (0.16 A_{090})^2 + 0.005^2, \quad (24)$$

$$\sigma_{E(B-V)} = \sigma_{A_{090}}/1.4156, \quad (25)$$

$$\sigma_{E(090-150)} = (1.4156 - 0.6021) \sigma_{E(B-V)}. \quad (26)$$

The individual Paper IV modulus error already contains its A_{090} term. We remove that contribution, $\sigma_{\mu,\text{no red}}^2 = \sigma_{\mu,\text{IV}}^2 - \sigma_{A_{090}}^2$, and then restore the single correct shared-reddening combination. The error of an absolute calibration point is

$$\sigma_{\bar{M},i}^2 = \sigma_{\bar{m},\text{int}}^2 + \sigma_{\mu,\text{no red}}^2 + \sigma_{E(090-150)}^2. \quad (27)$$

The same reddening moves corrected color and absolute SBF in opposite directions, giving

$$\text{Cov}(C_0, \bar{M}_{150,0}) = -\sigma_{E(090-150)}^2. \quad (28)$$

The photometric color error in each ring is the standard error of the usable pixel-color distribution after WCS registration and PSF matching. The two ring errors are combined as $\sigma_{C,\text{phot}}^2 = \sum_r w_r^2 \sigma_{C,r}^2$. The horizontal error used in the diagnostic linear regression is $\sigma_C^2 = \sigma_{C,\text{phot}}^2 + \sigma_{E(090-150)}^2$. This is a photometric error; the observed inner–outer color difference remains a separate radial-gradient diagnostic. The standard error does not claim to include every image-wide F090W/F150W calibration or PSF-matching systematic. Adding a deliberately larger 0.01-mag independent color floor leaves the linear LOO RMS unchanged at the reported precision, so this omission cannot drive the present conclusion that the color term is not required.

4.2 Zero-point calibration and predictive distance errors

The adopted calibration is color independent,

$$\bar{M}_{150,0} = a, \quad \mu_{\text{SBF}} = \bar{m}_{150,0} - a. \quad (29)$$

Consequently, neither the measured color nor its uncertainty enters a primary distance. A color law $f(C)$ is fitted only as a model-comparison diagnostic.

For a color law $f(C)$, the likelihood uses the effective variance

$$V_i = \sigma_{\bar{M},i}^2 + [f'(C_i) \sigma_{C,i}]^2 - 2f'(C_i) \text{Cov}(C_i, \bar{M}_i) + \sigma_{\text{int}}^2. \quad (30)$$

and maximizes the Gaussian likelihood including $\ln V_i$. For the adopted constant relation, $f'(C) = 0$, so all color and color-covariance terms vanish. The intrinsic

term σ_{int} is therefore fitted, not chosen to force reduced $\chi^2 = 1$. Fits with no horizontal error, an additional 0.01-mag object-to-object floor, and the historical half-ring proxy are reported as sensitivity tests.

For a new target under the adopted constant calibration, the target-measurement component is simply

$$\sigma_{\text{target,const}}^2 = \sigma_{\bar{m},\text{int}}^2 + \sigma_{A_{150}}^2. \quad (31)$$

For completeness, the corresponding expression used only in the diagnostic color-law fit is

$$\sigma_{\text{target}}^2 = \sigma_{\bar{m},\text{int}}^2 + [f'(C)\sigma_{C,\text{phot}}]^2 + \sigma_{A_{150}}^2 + [f'(C)\sigma_E]^2 - 2f'(C)\sigma_{A_{150}}\sigma_E, \quad (32)$$

where $\sigma_E \equiv \sigma_{E(090-150)}$ and $\text{Cov}(C_0, \bar{m}_{150,0}) = \sigma_{A_{150}}\sigma_E$ because both corrected observables move in the same direction when $E(B-V)$ changes. Its internal predictive variance then separates target measurement, population scatter, and finite calibration:

$$\sigma_{\mu,\text{int}}^2 = \sigma_{\text{target}}^2 + \sigma_{\text{int}}^2 + \sigma_{\text{cal}}^2. \quad (34)$$

The intrinsic term is the value fitted to the relevant 13-object LOO training sample. The finite-calibration term is the standard deviation of 300 non-parametric bootstrap predictions from those same 13 galaxies. The absolute distance error adds the common Paper IV TRGB/NGC 4258 zero point of 0.047 mag. A common F150W photometric zero point and a common finite-PSF-stamp shift cancel to first order between an apparent F150W SBF magnitude and an F150W absolute calibration measured with the same pipeline, so they are not added again as independent distance errors. They do affect the quoted absolute $\bar{M}_{150}$ zero point: the 0.047 mag TRGB, 0.0108 mag flux-scale, and 0.0167 mag PSF-stamp terms combine to 0.0510 mag. The common TRGB term does not enter the relative calibration fit.

Explicitly, a one per cent common F150W flux-scale uncertainty becomes $\sigma_{\text{flux}} = (2.5/\ln 10)(0.01) = 0.0108$ mag. The PSF-stamp term is the largest measured 129-versus-257-pixel spectral shift, 0.0167 mag. Therefore the common absolute-magnitude scale is

$$\sigma_{\bar{M},\text{common}} = \sqrt{0.047^2 + 0.0108^2 + 0.0167^2} = 0.0510 \text{ mag}. \quad (35)$$

For a same-filter SBF distance, the last two terms shift apparent and calibrated absolute F150W magnitudes together and cancel; only the 0.047-mag anchor is retained. Any modulus uncertainty is converted to a fractional distance uncertainty with

$$\frac{\sigma_D}{D} = \frac{\ln 10}{5} \sigma_{\mu}, \quad (36)$$

using the linear form for the quoted percentages (the exact asymmetric conversion is used in the machine-readable distance table).

For the primary SBF-minus-TRGB validation residual, the shared extinction term is $\sigma_{E(090-150)}$. The general color-law diagnostic gives $(1 + f')\sigma_{E(090-150)}$; this propagates shared reddening on the difference itself instead

of counting it independently in both axes. The validation variance is the quadrature sum of target measurement, target color photometry, this shared-reddening response, individual TRGB error without reddening, intrinsic scatter, and finite-calibration error. LOO uses the same calibrator set and is consequently an internal cross-validation, not an external test on unseen anchors.

The numerical contribution of each term is deliberately deferred to Section 6.4. Parentheses in the summary tables there give the full minimum–maximum range across galaxies rather than an uncertainty on the median.

5 Statistical framework and pre-specified experiment

The constant and linear calibrations are compared with the same Gaussian likelihood, fitted intrinsic scatter, AICc, and galaxy-level LOO prediction. The LOO experiment removes the target before fitting its calibration and is therefore an internal cross-validation of the 14 calibrators, not an external validation on independent galaxies.

The primary result uses all 14 galaxies, individual Paper IV thin-slice distances, no manual rejection, and one color-independent absolute F150W SBF zero point. The linear relation is a pre-specified test of whether a population correction is needed; quadratic and broken-line relations are exploratory. NGC 1399 remains in the primary fit; its constant-zero-point LOO residual is -0.143 mag, or 1.68σ with the validation uncertainty, and does not justify post-hoc removal.

Before inspecting the final distance residuals, the following sensitivity experiments were fixed: Paper IV thick-slice and Paper-III-style cluster anchors; constant and linear color laws; visually and automatically selected quality subsets; inner and outer annuli; $P_r = 0$, the adopted P_r , and $2P_r$; GCLF-width variants; PSF stamp size and detector position; the order and support used for residual winsorization; and an optional correlated-noise template. Nonlinear color laws and removal of individual galaxies remain exploratory and cannot replace the primary experiment.

6 Results

6.1 Extinction-corrected measurements

Table 1 is the single object-level audit table for the primary experiment. It joins the measured color and apparent SBF magnitude to the individual TRGB anchor and the resulting LOO residual; it is not an additional fit or a quality-selected subsample.

6.2 Color dependence: not required by the F150W data

We first ask whether color is needed before constructing the primary distance estimator. The color-independent

Table 1: Extinction-corrected GO-3055 measurements. The penultimate column is the color-independent LOO SBF-minus-TRGB residual. The printed “Virgo region” label denotes the broad environment class defined in Section 2.1. HQ IV reproduces the high-quality flag in Table 2 of Paper IV. The quoted $\sigma_{\overline{M}}$ contains individual errors but excludes common zero points. The annular half-difference is a radial diagnostic and is not included as random noise.

Galaxy	Environment	$C_{090-150,0}$ (mag)	$\overline{m}_{150,0}$ (mag)	$\overline{M}_{150,0}$ (mag)	$\sigma_{\overline{M}}$ (mag)	$ \Delta\overline{m} _{\text{ann}}$ (mag)	μ_{TRGB} (mag)	$\Delta\mu_{\text{LOO,const}}$ (mag)	HQ IV
NGC 1380	Fornax	0.510	28.087	-3.321	0.029	0.099	31.408	-0.122	+
NGC 1399	Fornax	0.570	28.157	-3.341	0.034	0.047	31.498	-0.143	—
NGC 1404	Fornax	0.559	28.177	-3.189	0.028	0.110	31.366	+0.020	+
NGC 1549	Other	0.533	27.930	-3.301	0.029	0.133	31.231	-0.101	—
NGC 3379	Other	0.557	27.012	-3.066	0.030	0.269	30.078	+0.152	+
NGC 4374	Virgo region	0.566	27.947	-3.165	0.031	0.094	31.112	+0.046	+
NGC 4406	Virgo region	0.616	27.849	-3.247	0.030	0.097	31.096	-0.042	+
NGC 4472	Virgo region	0.625	27.713	-3.303	0.035	0.029	31.016	-0.103	—
NGC 4486	Virgo region	0.652	27.808	-3.238	0.055	0.054	31.046	-0.033	—
NGC 4552	Virgo region	0.569	27.850	-3.073	0.030	0.175	30.923	+0.146	+
NGC 4621	Virgo region	0.565	27.688	-3.133	0.028	0.173	30.821	+0.081	+
NGC 4636	Virgo region	0.604	27.890	-3.180	0.031	0.110	31.070	+0.031	+
NGC 4649	Virgo region	0.623	27.873	-3.136	0.030	0.043	31.009	+0.077	—
NGC 4697	Virgo region	0.512	27.095	-3.229	0.032	0.173	30.324	-0.023	+

Table 2: Primary model comparison for all 14 galaxies. The linear relation tests whether a population correction improves on the adopted constant.

Model	a	b	σ_{int}	AICc / LOO RMS
Constant	-3.208	0.000	0.081	-23.75 / 0.0925
Linear	-3.209	0.224	0.080	-20.60 / 0.0981

fit gives

$$\begin{aligned} \overline{M}_{150,0} &= -3.208 \pm 0.024_{\text{stat}} \pm 0.051_{\text{common}} \text{ mag}, \\ \sigma_{\text{int}} &= 0.081 \text{ mag}. \end{aligned} \quad (37)$$

The statistical term is the standard deviation of 2000 galaxy-level bootstrap zero points. The common term combines the TRGB/NGC 4258 scale, the adopted F150W flux-scale allowance, and the finite-PSF-stamp term; only the TRGB term remains in a same-filter SBF distance. The bootstrap 16th–84th percentile interval of σ_{int} is 0.064–0.088 mag. At the median corrected color $C_0 = 0.5675$, the alternative linear fit is

$$\overline{M}_{150,0} = -3.2094 + 0.2244(C_{090-150,0} - 0.5675), \quad (38)$$

with intrinsic scatter 0.0798 mag. Bootstrap 16th–84th percentiles are $[-3.2313, -3.1815]$ mag for the intercept and $[-0.3044, 0.7136]$ for the slope, so the slope interval includes zero.

The constant model is better than the linear relation by $\Delta\text{AICc} = 3.15$ and also has lower LOO RMS. Color therefore does not improve the distance estimate in the present sample. This conclusion is not caused by the treatment of horizontal errors: the linear LOO RMS is 0.09809 mag with no color errors and 0.09810 mag either with the measured errors or after adding a 0.01-mag floor; using the old half-ring proxy gives 0.09847 mag. We consequently adopt Eq. 37; all subsequent primary distances and error budgets are color independent.

Quadratic and fixed-break broken-line relations were also tested only as exploratory shape checks. Their AICc/LOO-RMS pairs are $-19.77/0.0945$ and $-20.32/0.0922$ mag, respectively. The broken-line RMS is only 0.0003 mag below the constant-model value while its AICc is worse by 3.43. Neither model improves both criteria; their additional coefficients are therefore not promoted to a calibration table or a physical law.

The cleanest comparison with Paper III uses the same eight galaxies rather than comparing two different samples. For our F150W measurements versus $F090W - F150W$, the constant and linear models have LOO RMS values of 0.1014 and 0.1104 mag, respectively; the corresponding AICc values are -9.65 and -4.63 . For the Paper III F110W measurements of those same galaxies versus $g - z$, the linear model reduces the LOO RMS from 0.1114 to 0.0465 mag and has AICc -11.94 , compared with -7.97 for a constant. Thus the discrepancy is not produced merely by changing the galaxy list.

A crossed test of both SBF bands against both available colors constrains, but does not uniquely identify, the source of the discrepancy. On the same eight galaxies, F150W has Pearson $r = 0.28$ ($p = 0.504$) against $F090W - F150W$ and $r = 0.21$ ($p = 0.611$) against $g - z$. F110W gives $r = 0.82$ ($p = 0.0125$) and $r = 0.93$ ($p = 0.0010$), respectively. The two color indices themselves remain strongly correlated. Thus the F110W relation survives either color choice whereas the F150W relation does not. This points to the SBF passband as an important difference, but the eight objects still mix passband response, adopted distance prescription, environment, and reduction choices; the test does not isolate a unique physical cause. It also does not show that color is intrinsically useless or that no F150W color dependence can exist in a broader sample.

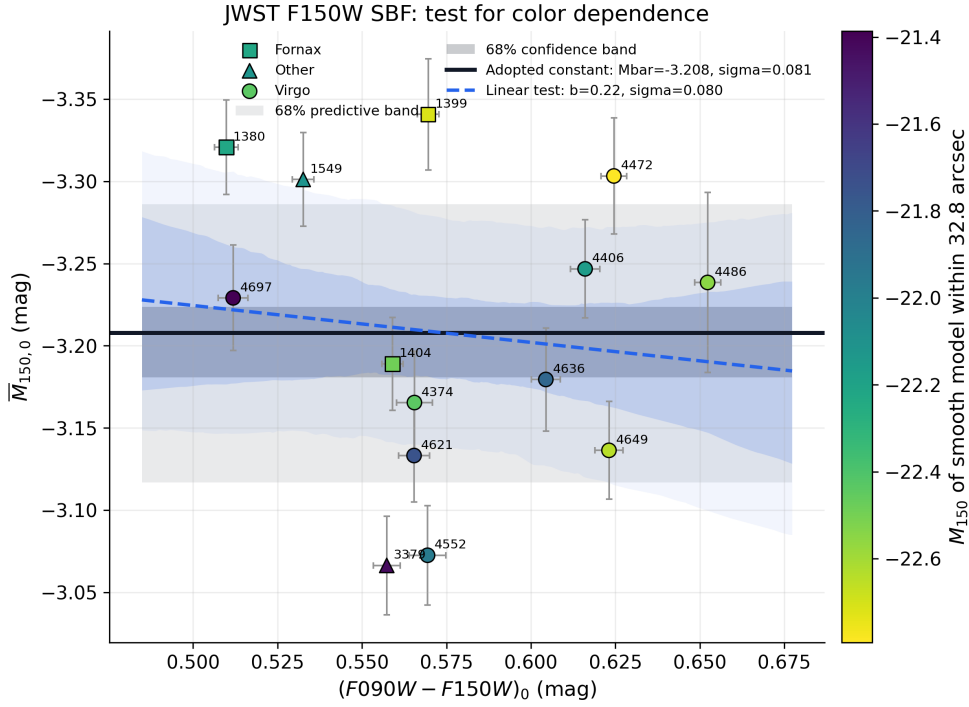


Figure 2: Extinction-corrected JWST F150W SBF calibration and the tested color dependence. The constant zero point is adopted; sloped and subset fits are diagnostics and do not define the primary distances. Marker shape gives environment and marker color the F150W aperture-luminosity proxy. For each displayed regression, the darker 68 per cent bootstrap band is the uncertainty of the mean relation; the lighter 68 per cent predictive band additionally includes the fitted intrinsic scatter.

6.3 Distance recovery and precision

The adopted color-independent relation gives LOO bias -0.0010 mag and RMS 0.0925 mag, corresponding to approximately 4.3 per cent scatter in distance. Its median predicted internal uncertainty is 0.0865 mag (4.1 per cent), and the median absolute uncertainty after the common zero point is 0.0984 mag (4.6 per cent).

The constant-zero-point validation pulls have RMS 1.08; eight of 14 objects lie within 1σ and all 14 within 2σ . Thus the error model is not obviously underestimated or inflated. The empirical LOO RMS and the median predicted uncertainty are distinct quantities and agree at the expected level for this small sample.

Table 3 gives the requested distance comparison in physical units. Its F150W column contains only the adopted color-independent LOO predictions: the listed galaxy was absent when its calibration was fitted. The rejected linear color law remains quantified in Table 2 and Figure 2, but is not presented as a second distance estimate. The Paper III F110W values are reconstructed for the eight overlapping objects from its Table 1 and Eq. (2), $\bar{M}_{110} = 1.86[(g - z)_{\text{PS}} - 1.30] - 2.760$. Their formal errors propagate the tabulated $\bar{m}_{110}$ and color errors and the published slope and intercept uncertainties as independent terms; common absolute-scale errors are excluded. The last column is the separate Jensen et al. (2015) F160W legacy comparison and has only five overlaps.

6.4 Error distribution by galaxy

Individual calibration-point errors are typically dominated by the TRGB anchor and power-spectrum measurement. NGC 4486 is exceptional because its background term is 0.0456 mag. In the predictive distance budget, intrinsic scatter is the largest component, followed by finite-calibration uncertainty; improving the formal P_0 error alone cannot reduce the total distance error below the present 4–5 per cent floor.

Table 4 summarizes the galaxy-dependent terms; Table 5 propagates them through an actual LOO prediction; and Table 6 lists coherent scale errors that must not be added as though they were independent for each galaxy. The largest median calibration-point component is the individual TRGB error. The exceptional 0.0456 -mag sky term belongs to NGC 4486.

6.5 Measurement-pipeline systematics

The local STPSF library contains five unit-normalized models per galaxy. Four offsets around the nominal detector position change the fitted SBF magnitude by at most 0.0043 mag. Stamp size matters more: relative to the 129-pixel model, truncation to 97, 65, and 33 pixels can shift the fitted amplitude by as much as 0.0127 , 0.0433 , and 0.1248 mag, respectively. More importantly, independently generated 257-pixel STPSFs show that the central 129 pixels miss 0.69–0.77 per cent of the 257-pixel flux and imply a common spectral shift of 0.0151 – 0.0167 mag over the fitted k range. We retain 0.0167 mag as a common absolute- $\bar{M}_{150}$ systematic. Including it

Which ingredient removes the color relation?

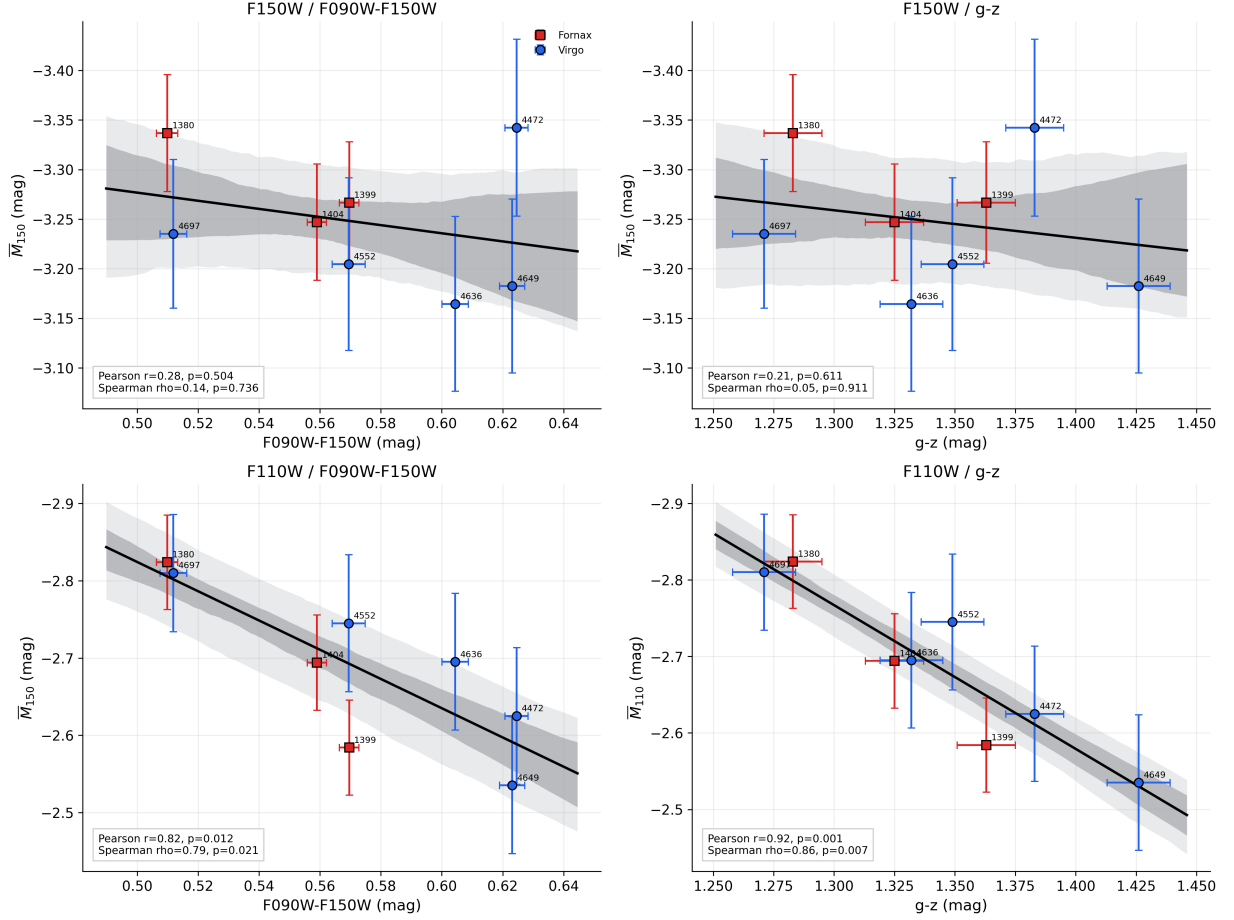


Figure 3: Crossed color test for the eight galaxies common to this work and Paper III. Each SBF band is tested against both $F090W - F150W$ and $g - z$. $F110W$ retains a strong relation with either population tracer, whereas $F150W$ does not; changing only the color index therefore does not recover a useful $F150W$ color correction. In every panel the darker 68 per cent bootstrap band is the confidence interval for the mean relation and the lighter 68 per cent band includes the empirical residual scatter of the corresponding fit.

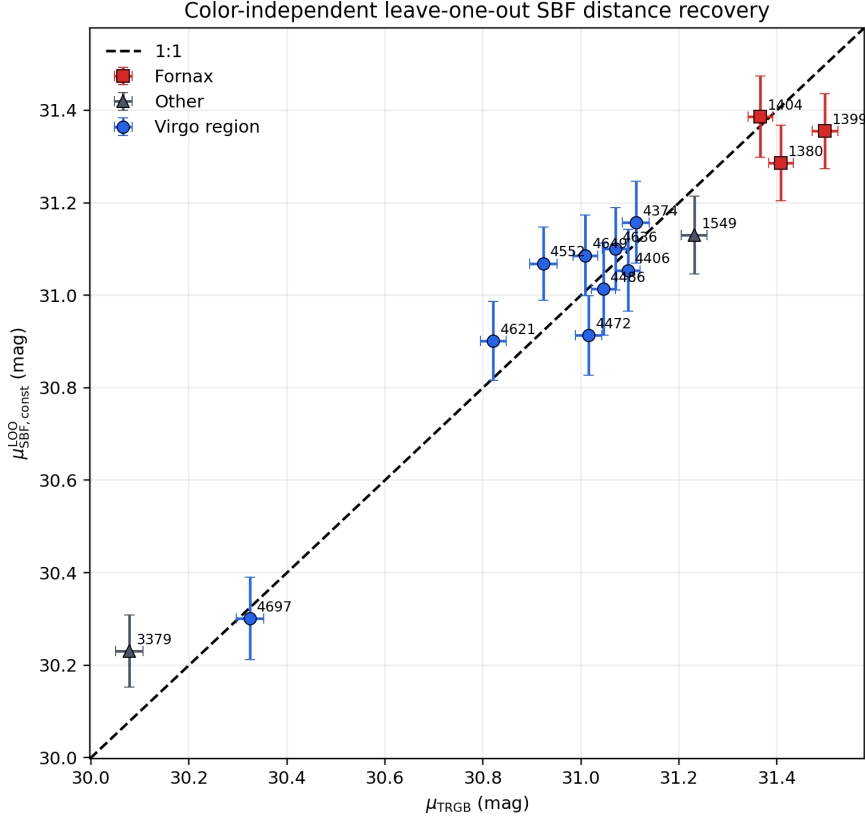


Figure 4: Color-independent LOO F150W SBF distance moduli compared with the individual Paper IV thin-slice TRGB moduli. Each target is excluded from its own calibration. Error bars are internal; the common absolute zero point is not shown.

Table 3: Distances in Mpc. TRGB IV denotes the individual Paper IV thin-slice anchor. “SBF adopted” is the color-independent JWST F150W LOO result. The rejected linear color-law distances are deliberately omitted from this main results table and retained only in the model-comparison diagnostics. Jensen III is the Paper III HST F110W reconstruction; Jensen 2015 is the independently labelled HST F160W legacy result. Ellipses mean that the galaxy is absent from that comparison sample. Quoted errors exclude common absolute zero points.

Galaxy	Env.	TRGB IV	SBF adopted	Jensen III F110W	Jensen 2015 F160W
NGC 1380	Fornax	19.12 ± 0.22	18.08 ± 0.68	18.98 ± 0.31	20.12 ± 1.19
NGC 1399	Fornax	19.93 ± 0.24	18.66 ± 0.70	19.80 ± 0.34	20.30 ± 1.19
NGC 1404	Fornax	18.76 ± 0.22	18.94 ± 0.77	19.44 ± 0.33	19.84 ± 1.22
NGC 1549	Other	17.63 ± 0.21	16.83 ± 0.66	...	...
NGC 3379	Other	10.37 ± 0.13	11.12 ± 0.40	...	...
NGC 4374	Virgo region	16.69 ± 0.21	17.04 ± 0.69	...	...
NGC 4406	Virgo region	16.57 ± 0.19	16.25 ± 0.66	...	...
NGC 4472	Virgo region	15.97 ± 0.20	15.23 ± 0.60	16.11 ± 0.27	16.19 ± 0.98
NGC 4486	Virgo region	16.19 ± 0.19	15.95 ± 0.73	...	...
NGC 4552	Virgo region	15.30 ± 0.20	16.36 ± 0.60	15.70 ± 0.27	...
NGC 4621	Virgo region	14.59 ± 0.17	15.15 ± 0.60	...	...
NGC 4636	Virgo region	16.37 ± 0.20	16.60 ± 0.68	16.30 ± 0.27	...
NGC 4649	Virgo region	15.91 ± 0.18	16.49 ± 0.66	16.19 ± 0.31	15.86 ± 0.98
NGC 4697	Virgo region	11.61 ± 0.15	11.49 ± 0.47	11.66 ± 0.19	...

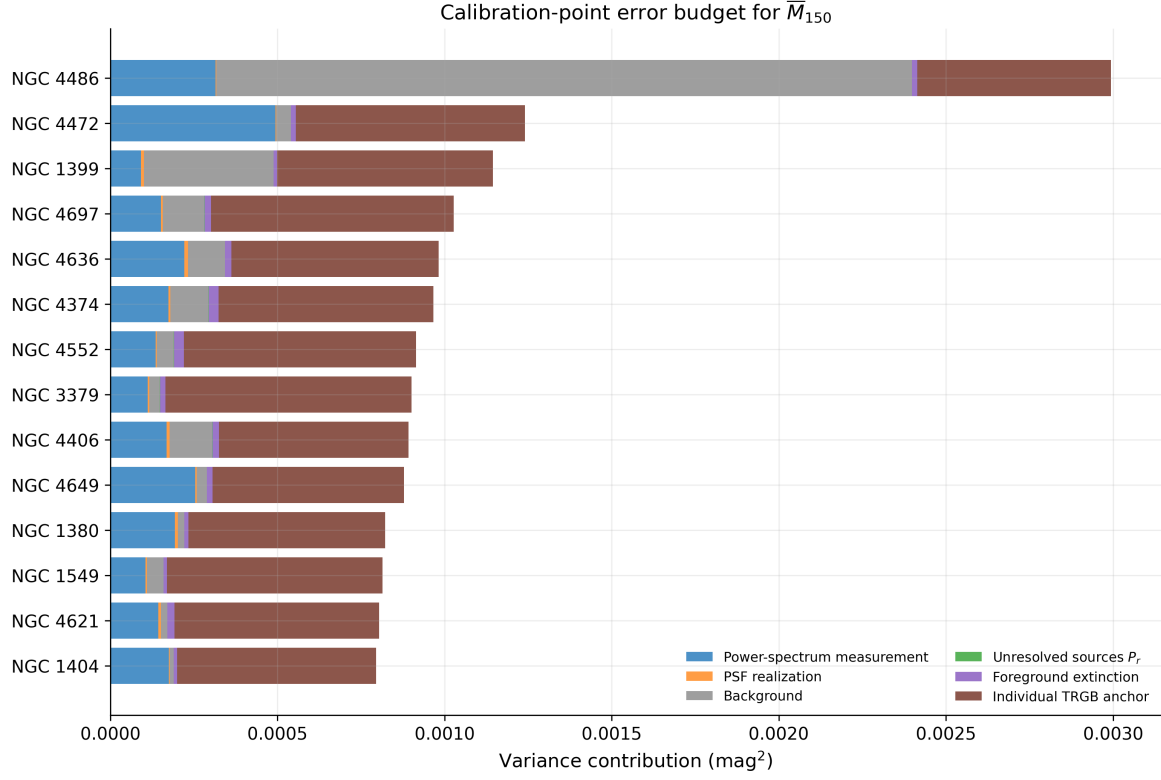


Figure 5: Variance budget of each absolute F150W calibration point. Stacked segments are the contributions to $\sigma_{\bar{M}}^2$ from the power-spectrum fit, sky, PSF realization, unresolved-source correction, foreground extinction, and individual TRGB anchor. The horizontal scale is variance in mag^2 , not a magnitude error; the total 1σ uncertainty is the square root of the complete bar.

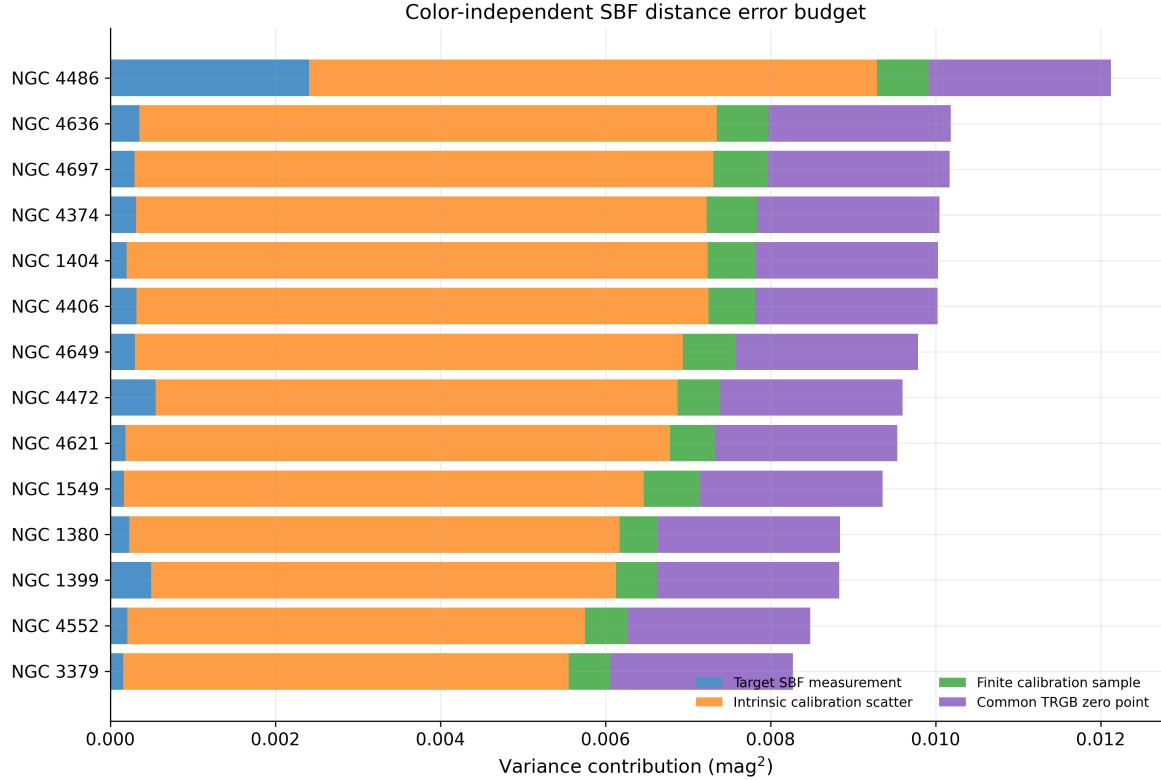


Figure 6: Variance budget of the adopted color-independent LOO SBF distance for each galaxy. The stacked terms are target SBF measurement, fitted intrinsic scatter, and finite-calibration uncertainty. The common TRGB scale is excluded from the internal bar and is added only for an absolute distance. As in Fig. 5, bar lengths are variances and must be square-rooted to obtain the quoted 1σ errors in magnitudes.

Table 4: Galaxy-dependent statistical uncertainties in magnitudes. Values are the median and, in parentheses, the minimum–maximum range across the 14 calibration points. Intrinsic and finite-calibration terms apply to predictions rather than to a single measured calibration point.

Component	Median	Range
Power-spectrum fit and k-window	0.0131	0.0096–0.0222
STPSF realization scatter	0.0022	0.0012–0.0033
Residual sources P_r	0.0006	0.0000–0.0013
Sky subtraction	0.0071	0.0035–0.0456
Paper IV TRGB (without reddening)	0.0251	0.0238–0.0271
Shared foreground reddening	0.0040	0.0031–0.0054
Total individual calibration point	0.0301	0.0282–0.0547

with the TRGB and NIRC*am* terms gives a 0.0510-mag absolute zero-point uncertainty; it does not change the internal LOO residuals.

The production size is 129 pixels, not 127. The odd dimension provides an unambiguous central sample and 64 pixels on either side. It retains 99.23–99.31 per cent of the corresponding 257-pixel STPSF flux for the three tested targets while reducing the repeated Monte Carlo convolution cost. The 257-pixel experiment shows that this choice produces a nearly coherent 0.015–0.017-mag absolute shift, rather than an uncontrolled object-to-object scatter. Retaining 129 pixels is therefore justified for the present same-pipeline distance comparison provided that the measured 0.0167-mag term is carried on the absolute $\overline{M}_{150}$ scale; a future sub-percent absolute calibration should instead use the larger stamp.

An empirical check was attempted with six bright, isolated compact sources in the NGC 3379 i2d frame. Their median stack is broader than STPSF and gives a formal 0.288-mag spectral difference. This number is *not* adopted as a PSF correction: the objects are not confirmed single stars, the i2d product has no DQ extension, and residual host/background structure broadens the stack. Thus the model-internal finite-stamp behavior is constrained, but an absolute empirical PSF zero point remains open until an isolated classified star is available. The detailed PSF diagnostics are shown in Appendix A; they are numerical validation plots, not additional astrophysical measurements.

The adopted 3.5σ residual winsorization was checked for all 14 galaxies with the galaxy model, catalogue mask, annular windows, PSFs, $E(k)$, and P_r held fixed. Four spectral branches were compared: no winsorization; winsorization of the raw residual before normalization; winsorization after normalization with a threshold estimated over the complete valid positive-model support; and the same operation with a threshold estimated only from the union of the two annuli. The third prescription is adopted because its threshold is defined in the same normalized variable that is Fourier analysed, while still producing a reusable full-frame FITS before the annuli are selected.

Relative to the former raw-residual prescription, the adopted full-support branch changes the weighted $\overline{m}_{150}$ by a median of -0.02954 mag over the 14 galaxies. The largest absolute change is 0.08798 mag for NGC 4649.

The ordering of these operations is consequently part of the final measurement definition, not a negligible perturbation and not an extra random error added after the fact. This winsorization remains an explicit departure from the Jensen pipeline; all alternative branches are retained as a sensitivity experiment rather than mixed into the primary calibration.

An extended model $P(k) = P_0E(k) + P_nN(k)$ was tested using an empirical correlated-noise spectrum. Over the available frequency interval, the $E(k)$ – $N(k)$ template correlations are 0.982–0.997. The additional coefficient therefore makes P_0 unstable and, in some fits, unphysical. We retain the Jensen-style $P_0E(k) + P_1$ model. A negative P_1 is allowed as the local intercept of a misspecified correlated-noise continuum; it is not interpreted as negative physical noise power.

The unresolved-source correction is negligible at the present precision. Across all galaxies, replacing the adopted P_r by either zero or $2P_r$ changes an individual weighted SBF magnitude by at most 0.0053 mag within a fixed residual measurement. This is much smaller than the 0.0925-mag cross-validated scatter of the adopted calibration. Re-fitting NGC 1399 and NGC 4472 with the Jensen-specific GCLF width of 1.35 mag changes any weighted magnitude by at most 0.00046 mag. Artificial-source completeness remains desirable, but it cannot alter the present calibration at a scientifically relevant level unless the current P_r estimate is wrong by far more than a factor of two.

A synthetic FFT-stage test passes a known normalized SBF field, mask, PSF, and white-noise term through the FFT and $P_0E(k) + P_1$ fit. Over 32 realizations the recovered magnitude has mean bias 0.0078 ± 0.0067 mag (standard error of the mean), consistent with zero; the trial-to-trial RMS is 0.0382 mag. The test validates the core unit conversion and spectral normalization, but not isophotal modelling, source detection, or the correlated i2d noise. The P_r and FFT-stage diagnostic plots are also collected in Appendix A.

A separate branch-recovery experiment isolates the order of normalization and winsorization. Sixty-four independent synthetic fields per annulus use a known P_0 , a radially varying positive galaxy model, the saved 129-pixel PSF, catalogue-like holes, and the same four spectral branches. Relative to the no-winsorization result for the same realization, the adopted full-support branch shifts the recovered magnitude by 0.00104 ± 0.00008 mag in the inner annulus and 0.00098 ± 0.00005 mag in the outer annulus (mean and standard error). Thus the chosen ordering does not introduce a detectable additional bias for an injected Gaussian SBF field. This is deliberately not described as a full end-to-end validation: it does not reproduce sky subtraction, isophotal modelling, source detection, unresolved-source completeness, or the non-Gaussian residual structure of the real mosaics.

6.6 Distance-anchor sensitivity

Table 7 compares the pre-specified thin-slice analysis, the Paper IV preferred thick-slice values, and a Paper-III-style experiment. The latter assigns the published Paper III means $\mu_{\text{Fornax}} = 31.424 \pm 0.057$ mag and $\mu_{\text{Virgo}} =$

Table 5: Decomposition of a predicted distance-modulus uncertainty. Entries are median (minimum–maximum) in magnitudes across the 14 LOO targets. Only the adopted color-independent calibration is shown. “Internal” combines target measurement, extinction, intrinsic scatter, and finite calibration. “Absolute” adds the common 0.047-mag TRGB scale. The final validation row also includes the individual target TRGB error and is not an uncertainty for a new SBF-only distance. Linear-model terms remain available only in the machine-readable sensitivity table.

Component	Adopted constant
Apparent SBF measurement	0.0169 (0.0122–0.0490)
Foreground-extinction propagation	0.0030 (0.0023–0.0040)
Fitted intrinsic population scatter	0.0814 (0.0734–0.0839)
Finite 13-galaxy LOO calibration	0.0240 (0.0215–0.0262)
Total internal SBF distance	0.0865 (0.0778–0.0996)
Total including common TRGB scale	0.0984 (0.0909–0.1101)
SBF–TRGB validation residual	0.0901 (0.0825–0.1025)

Table 6: Common systematic uncertainties. The combined absolute $\bar{M}_{150}$ scale is the quadrature of the preceding calibration terms. The same-filter distance row is a separate cancellation case and is not added to that total. The 1 per cent F150W flux-scale entry is a conservative common analysis allowance, not an uncertainty measured from these 14 galaxies.

Component	σ (mag)	Use
TRGB/NGC 4258 zero point	0.0470	absolute calibration
NIRCam F150W flux scale	0.0108	absolute calibration
STPSF finite-stamp term	0.0167	absolute calibration
Combined absolute $\bar{M}_{150}$ scale	0.0510	absolute calibration
Same-filter SBF distance scale	0.0470	F150W SBF distances

31.055 ± 0.086 mag to cluster members; NGC 1549, NGC 3379, and foreground NGC 4697 retain individual anchors.

For the adopted constant calibration, the thin/thick choice changes the zero point by only 0.006 mag and the LOO RMS by less than 0.001 mag. Cluster means reduce the fitted intrinsic scatter, as expected when depth and individual TRGB noise are suppressed, yet the gain against individual thin-slice distances is small. No conclusion depends on selecting the thin variant over the Paper IV preferred thick estimator. For the thin, thick, and cluster-mean anchors, the diagnostic linear-model LOO RMS values against their own anchors are 0.0981, 0.0994, and 0.0961 mag, respectively; none improves on the corresponding constant result.

6.7 Independent-band comparison with legacy F160W SBF

Five galaxies overlap the independently reduced Jensen et al. (2015) F160W sample. Their weighted mean modulus difference is $\mu_{150} - \mu_{160} = -0.117 \pm 0.070$ mag and the weighted RMS is 0.107 mag. The rank correlation between the two absolute SBF measurements is suggestive (Spearman $\rho = 0.87$, $p = 0.054$), but the

ordinary Pearson test is not significant ($r = 0.65$, $p = 0.233$). A fluctuation-color slope is likewise not detected. With only five common galaxies, this is a useful cross-band consistency check and a possible small offset to monitor, not an independent validation of the F150W distance scale. The object-level comparison is shown in Appendix B.

6.8 Annular behavior and environment

The inner annuli give slope 0.98 and intrinsic scatter 0.051 mag; the outer annuli give slope 0.07 and scatter 0.091 mag. The two points per galaxy are correlated and are not 28 independent calibrators. NGC 3379 has an exceptionally uniform residual image but a 0.269 mag annular SBF difference, demonstrating why the annular difference cannot be treated automatically as random measurement noise.

For the eight galaxies shared with Paper III, the direct F150W–F110W comparison is also substantially more coherent in the inner annulus (Pearson $r = 0.74$, $p = 0.035$; slope 0.56 ± 0.21) than in the outer annulus ($r = 0.31$, $p = 0.448$; slope 0.29 ± 0.36). In this population-sensitive sense, the inner measurements are more stable: they show the clearer cross-filter ordering and lower inter-galaxy scatter. This is not a blanket statement that the inner ring is numerically more robust. The all-14 spectral branch comparison gives the opposite result for sensitivity to high-amplitude residuals: the outer rings change less when the normalization and winsorization prescription is varied. The two notions of “stability” must not be conflated, and the present results do not support an order-of-magnitude precision claim for either ring.

Both circular rings are retained and combined because reproducing the radial geometry of Jensen/Paper III is a primary design constraint. They do not represent a claim that fixed angular annuli sample identical physical populations, nor that the outer ring is intrinsically preferable. A future optimization may choose physical or

Table 7: Sensitivity of the adopted constant calibration to the TRGB distance prescription. The final column always evaluates LOO predictions against the primary individual Paper IV thin-slice distances. Cluster-mean LOO values are not an independent galaxy-by-galaxy validation because different galaxies share an anchor.

Anchor	a	σ_{int}	LOO RMS vs anchor	LOO RMS vs thin
Paper IV thin	-3.208	0.081	0.0925	0.0925
Paper IV thick	-3.202	0.081	0.0932	0.0927
Paper III cluster means	-3.231	0.070	0.0924	0.0949

elliptical apertures, but that would be a new calibration experiment rather than a literal replication.

For the strict Virgo/Fornax subset, the seven Virgo members are NGC 4374, NGC 4406, NGC 4472, NGC 4486, NGC 4552, NGC 4621, and NGC 4649; the three Fornax members are NGC 1380, NGC 1399, and NGC 1404. A fitted Fornax shift is -0.157 mag. However, an exact permutation test gives $p = 0.150$, color and cluster membership are strongly confounded, and the step model is worse by $\Delta\text{AICc} = 0.59$. The step is a diagnostic, not a detection. Foreground extinction does not remove this pattern.

6.9 Structural-quality sensitivity

Quality flags were declared independently of the calibration fit. NGC 4486 contains the jet and strong large-scale structure; NGC 4472, NGC 4636, and NGC 4649 have strong residual structure; and NGC 1380 and NGC 4697 have incomplete outer model coverage. The primary sample nevertheless remains 14/14. Restricting it to the 11 visually acceptable or 10 automatically clean objects does not improve cross-validation: the constant-model LOO RMS becomes 0.0982 and 0.1036 mag, respectively, compared with 0.0925 mag for all 14. The nine Paper IV high-quality objects give 0.0869 mag; their linear model is worse at 0.1037 mag and therefore does not restore a useful color term. The test supports retaining the full sample while reporting the flags rather than using image appearance for post-hoc rejection.

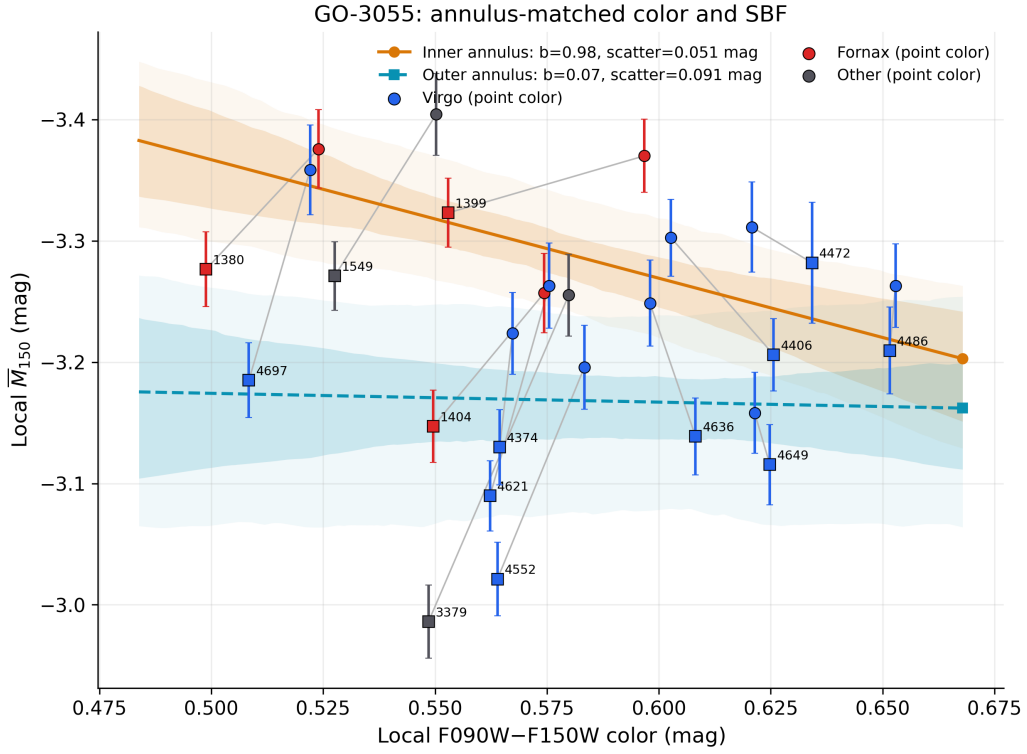


Figure 7: Local color and absolute F150W SBF magnitude in matched annuli. Circles and the solid orange line denote inner-annulus measurements and their fit; squares and the dashed cyan line denote the outer annulus. Point color encodes environment. Grey segments connect the two measurements of each galaxy. For each annular regression, the darker 68 per cent bootstrap band shows uncertainty in the mean relation and the lighter 68 per cent predictive band also includes intrinsic scatter.

7 Discussion

7.1 Accuracy of the method

The central empirical result is a cross-validated F150W SBF precision of approximately 4.3 per cent in distance for this 14-galaxy sample using one color-independent F150W zero point. The tested color correction does not improve this result and is not used in the adopted distances.

The uncertainty model passes a useful internal calibration check: the LOO pull RMS is close to unity and no target exceeds 2σ . This does not prove the absence of shared pipeline systematics. LOO galaxies reuse the same instrument, reduction, PSF construction, and TRGB scale; only a separate galaxy sample can provide an external accuracy test.

7.2 Remaining limitations

Three measurement systematics remain physically important. First, i2d noise is correlated. The attempted empirical $N(k)$ component is too degenerate with $E(k)$ over the fitted interval to stabilize a third coefficient, so the failure of that extension is documented rather than hidden behind a more complicated fit. Second, the retained 3.5σ winsorization is applied only after normalization, with one object-specific threshold estimated from the complete valid positive-model support. The all-galaxy branch comparison shows that changing the order relative to the former raw-residual prescription is

measurable, even though the synthetic branch-recovery test finds no additional bias for an injected Gaussian field. The selected threshold and ordering are therefore fixed methodological choices, not negligible corrections. Third, the empirical PSF comparison is not closed by a confirmed isolated star. The 129–257-pixel model comparison does, however, bound the current finite-stamp shape effect. Compact-source completeness and the P_r model remain approximate; the explicit $0-2P_r$ test bounds their impact on this calibration without pretending that completeness itself has been measured.

8 Conclusions

1. All 14 GO-3055 galaxies are retained in the primary analysis with individual Paper IV thin-slice TRGB anchors and no manual exclusions.
2. The calibrated constant is $\bar{M}_{150,0} = -3.208 \pm 0.024_{\text{stat}} \pm 0.051_{\text{common}}$ mag. It recovers the TRGB distances with 0.0925 mag LOO RMS (4.3 per cent) and negligible mean bias.
3. The median predicted color-independent precision is 4.1 per cent internally and 4.6 per cent after the common TRGB zero point. The observed pull distribution is consistent with the adopted errors.
4. The data do not require a color dependence: a linear color law worsens the LOO RMS to 0.0981 mag, its AICc is worse by 3.15, and its slope interval includes

zero. It is therefore retained only as a diagnostic and is not used for the primary distances.

5. Paper IV thick-slice distances and Paper III cluster means do not materially change the recovered accuracy. Thin slices remain primary only because that experiment was pre-specified, not because they give a better answer.
6. The Virgo–Fornax step and NGC 1399 exclusion remain diagnostics; neither is promoted to a detected effect or used to define the main sample.
7. The P_r sensitivity and synthetic FFT tests are negligible at the current distance precision. The 129-pixel PSF stamp instead introduces a 0.015–0.017 mag common absolute-calibration term.
8. The inner annulus has stronger cross-filter and population coherence, whereas the outer annulus is less sensitive to winsorization. Both are retained to reproduce the Jensen geometry; neither is declared universally superior.
9. The result is an internal LOO cross-validation. The $N(k)$ extension is rejected because its template is nearly degenerate with $E(k)$, and the fixed 3.5σ winsorization remains a measurable methodological choice. A confirmed empirical stellar PSF and an independent galaxy sample are required before claiming external accuracy.

9 Data and code availability

The calibrated JWST observations are public through MAST under GO-3055; the archive collection is accessible through DOI [10.17909/z9ch-wk24](https://doi.org/10.17909/z9ch-wk24). No proprietary data are used. The machine-readable derived products required to reproduce the tables and figures are listed in Appendix B and retained with this working paper.

The measurement and analysis code is presently the version-controlled project copy described in Appendix C. A public archival release and permanent code DOI have not yet been assigned and must be added before submission. This statement does not claim that an unarchived local repository is publicly available.

A Measurement-pipeline diagnostic plots

The following plots document numerical sensitivity tests requested during pipeline validation. They support the systematic-error bounds quoted in the main text and do not enter the adopted color-independent distance estimator as additional data.

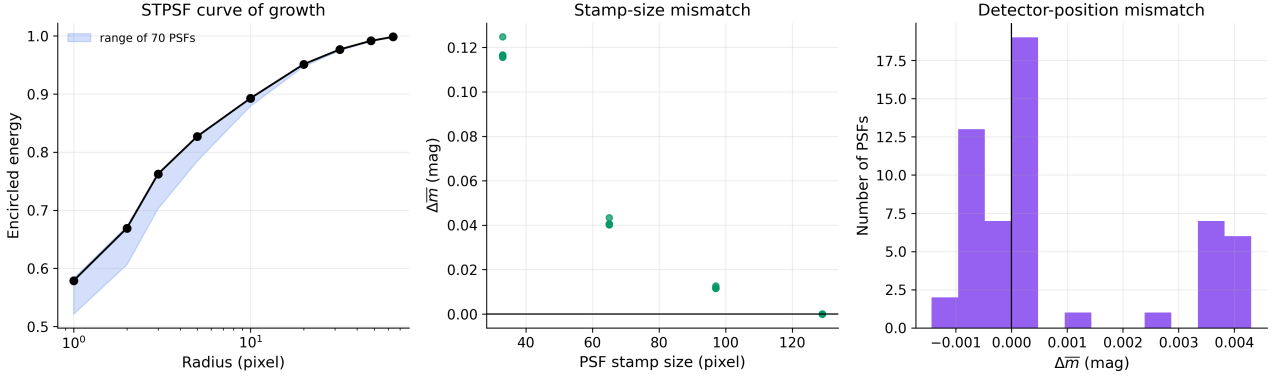


Figure 8: Internal PSF tests. Left: STPSF encircled-energy curves for all 70 models. Middle: spectral shifts caused by truncating the saved 129-pixel model. Right: shifts from the four detector-position variants.

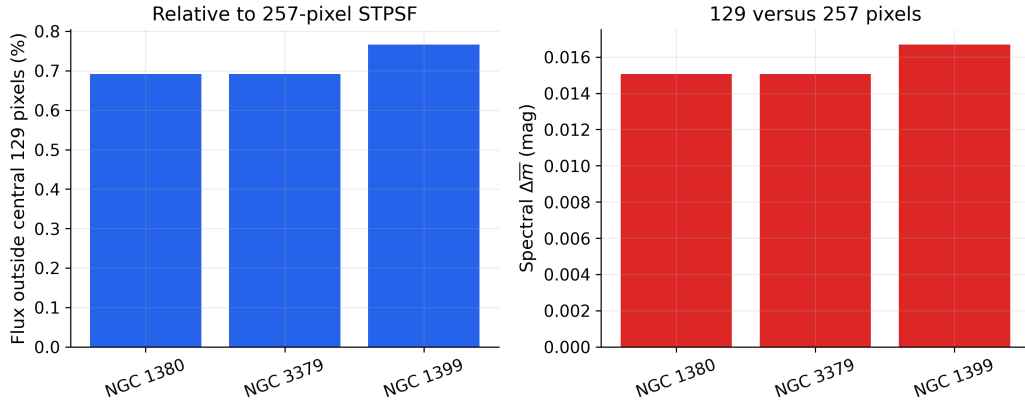


Figure 9: Absolute stamp-size test for three representative targets. The left panel gives flux outside the central 129 pixels relative to a 257-pixel STPSF; the right panel translates the shape mismatch over the fitted frequency range into an SBF-magnitude shift.

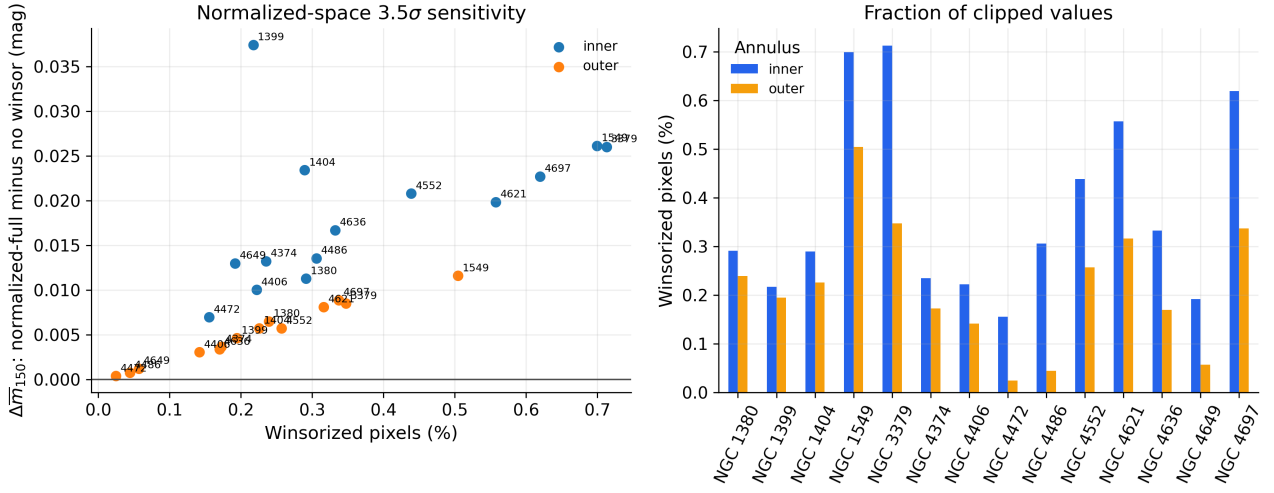


Figure 10: All-galaxy diagnostic for the adopted normalized-full 3.5σ winsorization. Left: change in ring SBF magnitude relative to no winsorization against the fraction of affected pixels. Right: affected-pixel fractions for the inner and outer annuli. These values come from the final `normalized_full_3p5` products and define the quantitative winsorization check used in this work.

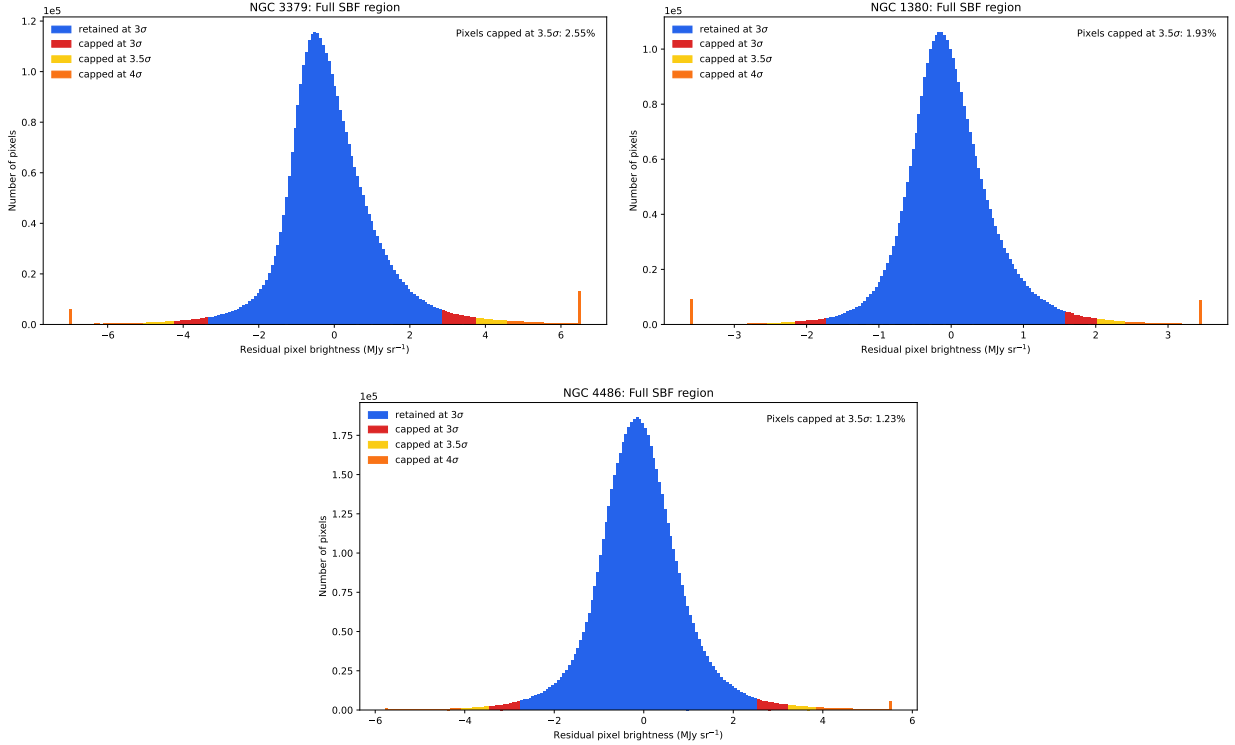


Figure 11: Legacy three-galaxy raw-residual pilot retained to show why the order of normalization and winsorization was revisited. The colored tails belong to that earlier experiment: they neither define the final thresholds nor give the adopted affected-pixel fractions. The quantitative final-branch check is Figure 10.

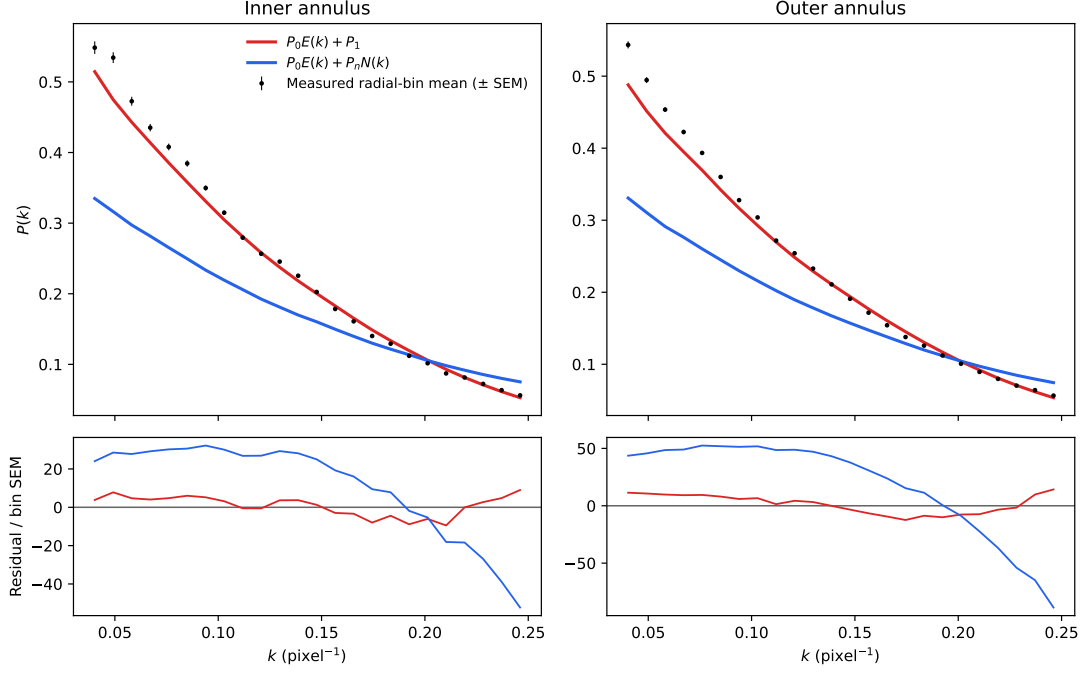


Figure 12: Legacy raw-residual pilot fits for representative inner and outer annuli. Black points are radial-bin means of the measured two-dimensional Fourier power; their error bars are standard errors over contributing Fourier modes. The red curve is the adopted $P_0E(k) + P_1$ model. The blue curve is the rejected $P_0E(k) + P_nN(k)$ diagnostic; its poorer behavior follows from the near degeneracy of $E(k)$ and $N(k)$, not from an additional detected noise component. Lower panels show residuals divided by the bin errors. This panel supports the retained functional form only; its numerical coefficients are not used by the adopted normalized-full branch.

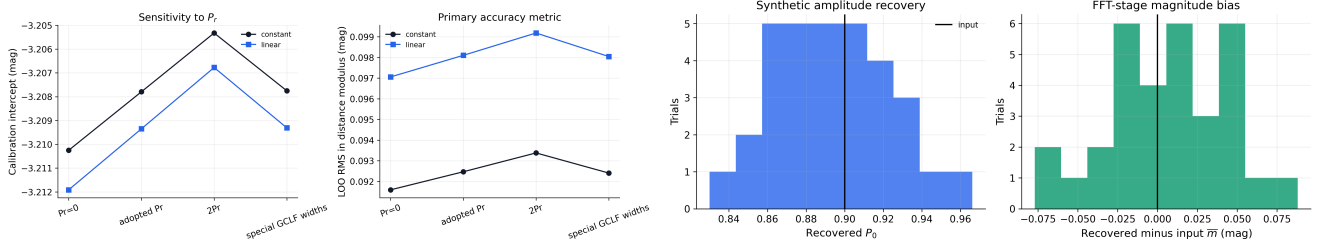


Figure 13: Left: calibration and color-independent LOO metric under the P_r sensitivity experiments. Right: recovery of known P_0 and SBF magnitude in the synthetic FFT-stage test.

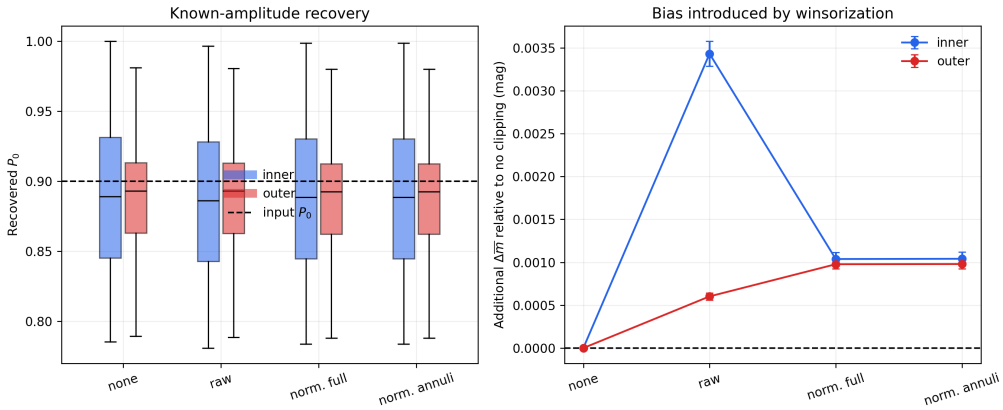


Figure 14: Synthetic recovery test for the order of normalization and winsorization. Each spectral branch is applied to the same injected SBF and white-noise realization before the inner and outer annuli are fitted. The comparison isolates this ordering only; it is not an end-to-end test of sky, isophotal, catalogue, or correlated-noise systematics.

B Auxiliary scientific and quality diagnostics

The following figures and table retain useful checks that do not define the primary calibration. They are separated from the main Results section so that the central claims are not obscured by exploratory host-luminosity and small-overlap comparisons.

The aperture-magnitude test detects no direct correlation for all 14 objects (Pearson $p = 0.089$), Virgo alone ($p = 0.452$), or the three Fornax objects ($p = 0.954$). After subtracting the rejected linear color relation, the corresponding diagnostic values are 0.053, 0.321, and 0.900. These numbers cannot be interpreted as a total-luminosity dependence because the fixed angular aperture itself is confounded with environment and distance.

B.1 Machine-readable products

The full distance table is `runs/sbf2_go3055/analysis/tables/go3055_distances_mpc.csv`. The joined TRGB/F150W/F110W/F160W comparison is `runs/sbf2_go3055/analysis/tables/go3055_distance_comparison_all_methods.csv`. The predictive errors are in `runs/sbf2_go3055/analysis/tables/go3055_distance_predictions_full_errors.csv`; the component table is `runs/sbf2_go3055/analysis/tables/go3055_error_budget.csv`; anchor tests are in `runs/sbf2_go3055/analysis/tables/go3055_distance_anchor_sensitivity.csv`.

Additional machine-readable systematics products are `runs/sbf2_go3055/analysis/tables/go3055_psf_129_vs_257_sensitivity.csv`, `runs/sbf2_go3055/analysis/tables/go3055_psf_detector_position_sensitivity.csv`, `runs/sbf2_go3055/analysis/tables/go3055_Pr_calibration_sensitivity.csv`, `runs/sbf2_go3055/analysis/tables/go3055_synthetic_fft_summary.csv`, and `runs/sbf2_go3055/analysis/tables/go3055_structural_quality_and_physical_annuli.csv`. The consolidated three-galaxy noise, winsorization, and PSF experiment is `runs/sbf2_systematics/pilot/tables/sbf2_three_galaxy_systematics_pilot.csv`; its adopted decisions are recorded separately in `runs/sbf2_systematics/pilot/tables/sbf2_three_galaxy_systematics_decision.csv`. The final all-14 normalization/winsorization comparison is retained under `runs/sbf2_normalized_winsor/batch/aggregates/`; its per-target status table is `runs/sbf2_normalized_winsor/batch/target_status.csv`, and the synthetic branch-recovery summaries are in `runs/sbf2_normalized_winsor/recovery/`. The compact uncertainty summaries printed in the article are also available as `runs/sbf2_go3055/analysis/tables/go3055_individual_error_summary.csv`, `runs/sbf2_go3055/analysis/tables/go3055_predictive_error_summary.csv`, and `runs/sbf2_go3055/analysis/tables/go3055_common_systematics.csv`.

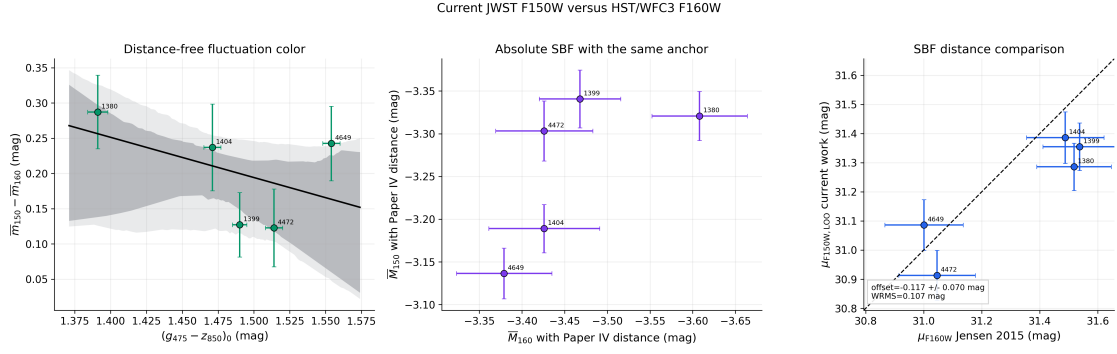


Figure 15: Comparison with the five independently reduced Jensen et al. (2015) F160W overlaps. The sample is too small for a new calibration claim; the plot is retained only as the separately labelled cross-band consistency check described in the main text. The black line in the left panel is the fitted fluctuation-color trend; the dashed black line in the right panel is equality. The middle panel has no fitted black line. Around the fitted relation in the left panel, the darker 68 per cent bootstrap band describes uncertainty in the mean and the lighter 68 per cent predictive band includes the empirical residual scatter of the fitted relation.

GO-3055: SBF and smooth-model magnitude within 32.8 arcsec

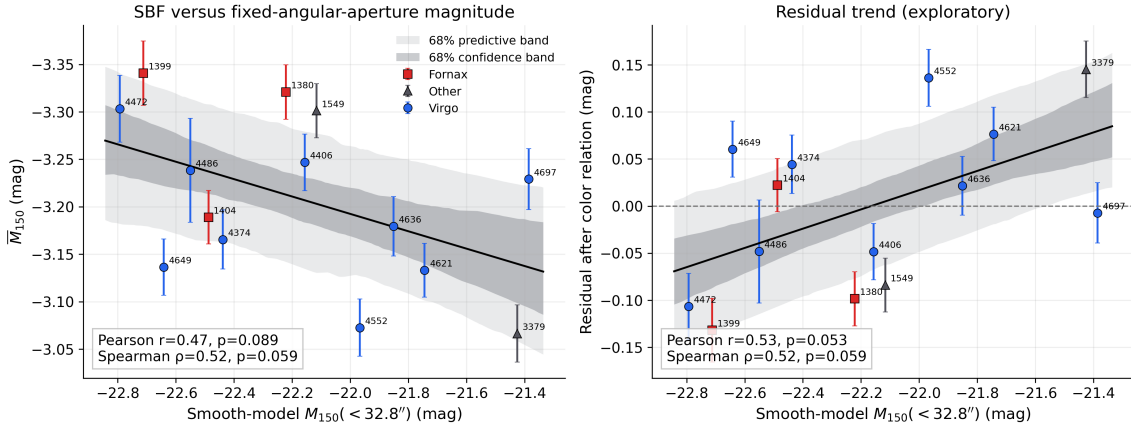


Figure 16: F150W SBF versus the smooth-model magnitude inside the fixed 32.8-arcsec aperture. This quantity is an aperture luminosity, not a total galaxy magnitude, and the same angular radius covers different physical radii. The right panel subtracts the rejected linear color relation only as an exploratory diagnostic; neither panel enters the adopted distance estimator. In both panels the darker 68 per cent bootstrap band is the confidence interval for the mean relation and the lighter 68 per cent predictive band also includes the empirical residual scatter of the fitted relation.

Table 8: Predeclared structural-quality diagnostics. The automatic flag is set at a structure score of 0.80; the visual score is on the user-defined 0–10 scale. These flags define sensitivity samples and do not remove any galaxy from the primary 14-object calibration.

Galaxy	Visual score	Auto score	Auto flag	Coverage flag	Predeclared issue
NGC 1380	7.00	0.40	No	Yes	incomplete outer model coverage
NGC 1399	9.50	0.05	No	No	
NGC 1404	8.00	0.12	No	No	
NGC 1549	9.00	0.18	No	No	
NGC 3379	10.00	0.26	No	No	
NGC 4374	8.00	0.15	No	No	
NGC 4406	8.00	0.64	No	No	
NGC 4472	4.00	1.61	Yes	No	strong large-scale structure
NGC 4486	3.00	1.47	Yes	No	jet residual; strong large-scale structure
NGC 4552	9.00	0.20	No	No	
NGC 4621	7.00	0.35	No	No	
NGC 4636	6.00	0.97	Yes	No	strong large-scale structure
NGC 4649	5.00	1.40	Yes	No	strong large-scale structure
NGC 4697	6.00	0.40	No	Yes	incomplete outer model coverage

C Reproducibility

The galaxy modelling and source-mask notebook is `code/sbf-2.ipynb`; its dedicated GO-3055 runner is `code/run_sbf_2_batch.py`. The final normalized-full spectral branch is implemented in `code/sbf2_normalized_winsor_core.py` and executed sequentially by `code/run_sbf_2_normalized_winsor.py`. It reuses the saved galaxy model, masks, PSFs, annular geometry, and cached $E(k)$ rather than repeating isophotal modelling. Exploratory fits and diagnostic figures are generated and displayed inline by `code/sbf-2-graph.ipynb`, which reads completed batch products and does not repeat the SBF measurement. The three-galaxy winsorization, noise, and 129-versus-257-pixel PSF diagnostics are generated by `code/sbf-2-systematics.ipynb`. The all-galaxy spectral-branch comparison is summarized in `code/sbf-2-normalized-winsor.ipynb`; its synthetic recovery is implemented by `code/sbf2_normalized_winsor_recovery.py`. The adopted constant-calibration tables, distance-recovery figures, error-budget figure, comparison tables, and provenance records are generated from those saved products by `code/build_sbf2_article_tables.py`.

The implementation uses NumPy, SciPy, Astropy, Photutils, Matplotlib, and STPSF [17–23]. Exact installed versions, rather than only literature citations, are given in Table 10.

The 28 input i2d headers are heterogeneous archive products rather than a uniform local reprocessing: 2 files record JWST pipeline 1.19.1, CRDS 12.1.11, and `jwst_1413.pmap`; 22 record pipeline 1.20.2, CRDS 13.0.6, and `jwst_1464.pmap`; and 4 F090W files record pipeline 2.0.1, CRDS 13.1.13, and `jwst_1535.pmap`. Exact counts are generated directly from the files used by the batch and retained in `go3055_input_calibration_provenance.csv`. This records the reduction state exactly, but does not prove that mixing contexts is harmless; uniform reprocessing is a residual reproducibility/systematics check. The base model-and-mask run records Git HEAD `5d0fd1496474...` and the exact executed `sbf-2.ipynb` SHA256 `a1828d2d8d97...`. That snapshot defines the galaxy model, masks, annuli, PSFs, and reusable $E(k)$ caches, but does not by itself define the adopted spectrum measurement. The latter is fixed jointly by the hashes of `sbf2_normalized_winsor_core.py` and `run_sbf_2_normalized_winsor.py`, the version/branch pair `sbf2-normalized-winsor-v3/normalized_full_3p5`, and the per-target experiment keys in the v3 result JSON files. The frozen F150W science-result commit is `d89e09482978...`. Full identifiers are retained in `go3055_software_provenance.csv`.

From the project root, the reproducible analysis commands are

```
export MPLBACKEND=Agg
export MPLCONFIGDIR=.matplotlib
astro_env/bin/python code/run_sbf_2_batch.py \
  --no-download --no-cleanup-inputs
astro_env/bin/python \
```

```
code/run_sbf_2_normalized_winsor.py \
  --stop-on-error
astro_env/bin/python \
  code/sbf2_normalized_winsor_recovery.py
astro_env/bin/jupyter nbconvert \
  --to notebook --execute --inplace \
  code/sbf-2-graph.ipynb \
  --ExecutePreprocessor.timeout=7200
astro_env/bin/jupyter nbconvert \
  --to notebook --execute --inplace \
  code/sbf-2-systematics.ipynb \
  --ExecutePreprocessor.timeout=7200
astro_env/bin/python \
  code/build_sbf2_article_tables.py
(cd texts/paper_work && latexmk -pdf \
  -outdir=build -interaction=nonstopmode \
  -halt-on-error go3055_jwst_sbf_article_draft.tex)
```

The Agg backend is mandatory for non-interactive execution on macOS; figures are saved and also displayed inline in the notebook.

Table 9: Fixed processing configuration for the reported run. Values not listed here remain recorded in the executed notebook snapshot.

Stage	Adopted configuration
Input and bands	JWST/NIRCam i2d; F150W SBF signal; F090W–F150W color; native pixel scale $\simeq 0.03123$ arcsec
Science regions	circular 8.2–16.4 and 16.4–32.8 arcsec annuli
Isophotes	initial semimajor axis 50 pixels; free ellipticity and position angle; 10-pixel sampling; maximum 2000 pixels; real-pixel fit with filled-gap fallback
Compact masks	primary 3.5σ , 9 connected pixels, 2-pixel dilation; residual catalogue 2.5σ , 4 connected pixels, deblending, 3-pixel dilation
Residual limiting	divide by $\sqrt{M}$ first; estimate one symmetric 3.5σ interval from the complete valid positive-model support; winsorize and save the full normalized FITS; select each ring and subtract its mean only immediately before the FFT. Raw, no-winsorization, and annulus-union branches are diagnostics only
PSF	STPSF; nearest time-matched WSS OPD; nominal plus four detector offsets; 129 pixels; 7 wavelengths; unit normalization
Fourier model	80 radial bins; 64 Monte Carlo realizations per $E(k)$ and PSF; $P_0E(k) + P_1$; primary $k = 0.04$ – 0.25 ; $k_{\min} = 0.01, 0.03$ sensitivity windows
Unresolved sources	Gaussian GCLF plus background-galaxy power law; 12-mag faint integration; 25 per cent P_r uncertainty; 0, P_r , and $2P_r$ tests
Calibration	all 14 primary; individual Paper IV thin-slice anchors; constant zero point with fitted intrinsic scatter; 300 LOO prediction bootstraps. The linear errors-in-variables fit with reddening covariance and 2000 fit bootstraps is diagnostic only

Table 10: Software and provenance identifiers. Long hashes are abbreviated in print and retained in full in the accompanying CSV.

Item	Version or identifier
Input i2d products (2/28)	pipeline 1.19.1; CRDS 12.1.11; jwst_1413.pmap
Input i2d products (22/28)	pipeline 1.20.2; CRDS 13.0.6; jwst_1464.pmap
Input i2d products (4/28)	pipeline 2.0.1; CRDS 13.1.13; jwst_1535.pmap
Python	3.13.2
numpy	2.3.5
astropy	7.2.0
photutils	2.3.0
stpsf	2.1.0
scipy	1.16.3
matplotlib	3.10.8
pandas	2.3.3
Adopted F150W SBF contract	sbf2-normalized-winsor-v3; nor- malized_full_3p5; kmin=0.04 33f9e794e329...
Normalized SBF core SHA256	2505a6de87dc...
Normalized SBF runner SHA256	5d0fd1496474...
Product-generation Git HEAD	a1828d2d8d97...
Executed sbf-2.ipynb SHA256	974aabcd497c...
Current sbf-2-graph.ipynb SHA256	32cfe6de01f8...
Current sbf-2-systematics.ipynb SHA256	0c8f75bf6657...
Current article TeX SHA256	3ff99f1272d8...
Article-table builder SHA256	d89e09482978...
F150W science-freeze Git commit	

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